



Nathaniel F. Barrett
INSTITUTE FOR THE
BIOCULTURAL STUDY OF
RELIGION,
UNITED STATES

Principles of Neurotheology

By Andrew B. Newberg

Surrey: Ashgate, 2010. 286 pp., hb. £ 55.00, pb. £ 16.99; ISBN: 978-1-4094-0810-9/ 978-0-7546-6994-4.

If you have read anything about brain-imaging studies of religious experience in the last ten years, it is likely that you have come across the name of Andrew Newberg. A leading expert in neuroradiology, he has conducted some of the highest-profile studies of the neurobiological correlates of experiences induced by meditation and prayer. Moreover, since co-authoring *The Mystical Mind* (1999) with Eugene d'Aquili, Newberg has established himself in the public eye with numerous mainstream media appearances and several provocatively titled books aimed at a general audience (e.g. *How God Changes Your Brain*, 2009). With *Principles of Neurotheology* (2010), Newberg seems to be turning toward a different audience: his academic peers in the sciences and humanities. His stated purpose is to clarify the premises, goals, and methods of a massive interdisciplinary project that he calls 'neurotheology'; his unstated purpose, I suspect, is to entice more scholars to join him in this unusual venture.

Newberg's vision for neurotheology is extraordinarily ambitious. Though his initial definition of neurotheology is deceptively simple—'the field of study linking the neurosciences with religion and theology' (p. 45)—as this linkage is further articulated, it becomes clear that Newberg seeks to do more than foster interdisciplinary dialogue. His goal is the integration of religious and scientific perspectives in pursuit of an overarching 'megatheology' rooted in 'two universal elements—religion and the human brain' (pp. 65-66). However, though he repeatedly calls for an equal and mutually respectful partnership between religious and scientific perspectives, by the end of the book it is clear that the real engine of neurotheology is brain science. Newberg hopes that brain science, when coupled with a rich phenomenology of religious experience, will allow us finally to tackle perennial questions about the nature of consciousness, the nature of reality, and the existence of God. The religions of the world may have already granted us experiential access to ultimate reality—Newberg is always careful to leave this possibility open—but he believes that brain science will play a pivotal role in developing a systematic understanding of religious experience, and hopes that this understanding will improve the human condition and guide us toward a 'new enlightenment' (p. 267).



Newberg builds his case for neurotheology through the exposition of fifty-four principles. Although he aspires to a set of ‘necessary’ principles that together constitute the ‘ideological foundations’ of neurotheology (pp. 1, 267), in the course of his argument he tends to offer the principles as suggestions or guidelines rather than necessary axioms. The principles range from commonsensical tenets of open inquiry (e.g. Principle III, p. 46) to some rather tendentious claims about the neural constraints of religious belief and experience (Principle XI, p. 64) and the ‘fundamental uncertainty’ of all beliefs about reality (Principle XXIX, p. 214). More than anything, it seems that Newberg’s use of principles is a rhetorical strategy designed for maximum clarity and straightforwardness, and in this respect he succeeds admirably: one could get the gist of his book by reading just the principles printed in bold.

However, while the principles are clear and straightforward, they are not adequately justified. In particular, Newberg does not sufficiently examine, let alone justify, the epistemological premises that make brain science so important for the synthesis of religion and science. And yet Newberg’s proposal that brain functions are a crucial tool for the interpretation of religious beliefs—what he calls ‘neurotheological hermeneutic’ (see Principle XVII, p. 84; Chapter 5: ‘Towards a Neurotheological Hermeneutic,’ pp. 87–114)—is the most distinctive feature of his methodological approach. Indeed, I would argue that Newberg’s picture of how the brain determines our experience of reality is not just the lynchpin of his planned synthesis of religious and scientific perspectives; it is the fundamental premise of ‘neurotheology.’

The importance of Newberg’s brain-centered epistemology comes into full view in the tenth and final chapter, ‘Epistemological Issues in Neurotheology’ (pp. 249–265), where he proposes that brain science, coupled with the phenomenology of ‘primary epistemic states,’ might provide leverage on questions about the ultimate nature of reality. Ironically, this leverage is gained by placing us in an epistemological predicament of seemingly hopeless isolation. Newberg argues that we ‘are trapped with our brain peering out into the world and reconstructing it the best we can’ (p. 252). From this position, our only indication of what is real is what *feels* real to us. By combining phenomenology with neurobiology, however, Newberg believes that we might be able to find our way out of our brains (pp. 262–265).

In my estimation, Newberg’s uncritical conflation of brain activity with cognition is the book’s major flaw. No one can doubt that the brain is the most important organ of human cognition, but that does not necessarily mean that we are trapped within the brain. In Newberg’s vision of neurotheology, the brain is so important that it crowds out all other objects of scientific investigation. This narrow scientific focus is reinforced on the religious side by a preoccupation with the phenomenology of religious experience. In his discussions of the relevant data for building a scientific understanding of religion, Newberg focuses almost exclusively on ‘subjective’ accounts of spiritual experience (p. 2) and makes no mention of physical, cultural, and social environments. My guess is that this narrow focus is justified by the assumption that all environmental factors must be ‘filtered’ by the brain (pp. 87–88), so that a full understanding of human

experience, religious and otherwise, must await a thorough investigation of these neural ‘filters’ (p. 64). At the heart of Newberg’s vision of neurotheology, then, is the unexamined notion that we are trapped within the worlds created for us by our brains, so that brain science is crucial to any attempt to understand what reality is really like ‘out there.’ This results in a highly truncated view of what is relevant to the scientific study of religion.

It should be noted that many of Newberg’s presuppositions about the brain and cognition are commonplace in cognitive science, where the standard epistemological stance seems to be a kind of ‘evolutionary Kantianism,’ i.e. a formalist and representationalist view bolstered by evolutionary biology and given specious clarity by the computational metaphor of the brain as an information processor. This view is so widely held that it may seem unfair to take Newberg to task for being insufficiently critical about its premises. But as a self-conscious attempt to establish the theoretical and methodological groundwork of a new field of research, *Principles of Neurotheology* is precisely the sort of work that should take pains to examine its premises. Newberg not only fails to address the many criticisms of brain-centered, representationalist epistemology, he shows no sign of even being aware that they exist. This is an egregious oversight given the prominence of ‘extended,’ ‘embodied,’ and ‘situated,’ theories of mind in recent years.¹

Another major flaw of the book is that it does not engage important predecessors and peers. Newberg only briefly acknowledges other proponents of what he calls ‘neurotheology’ (p. 12) and does not examine the history of the term’s usage. Although he does provide a helpful review of relevant scientific research (pp. 170-178), for the most part Newberg settles for extensive citations rather than detailed engagement with others’ views. If Newberg truly believes that neurotheology is essentially a collaborative endeavor, that conviction needs to be reflected in the way that he writes about the subject.

Newberg also fails to address important objections to neurotheology. Lurking in the background of Newberg’s project is the question of whether ‘immaterial’ or ‘supernatural’ entities can really be the objects of empirical inquiry. Of course this question cannot be pursued without more precise definitions. Despite his claim that ‘religious beliefs virtually always begin with a supernatural foundation’ (p. 51), Newberg does not define the supernatural, and never faces up to the questions begged by his insistence that neurotheology remain open to supernatural explanations. As far as I can tell, Newberg is not shielding a specific theological agenda; rather he is trying to remain true to his principles of open inquiry. But an open mind still must be clear about what it means for *any* entity to be susceptible to empirical investigation: these key principles of inquiry are missing from Newberg’s book. When Newberg ventures to speculate about what empirical evidence of an immaterial cause would look like, he provides what amounts to an inference from the *absence* of evidence: no detectable material cause, therefore... (see pp. 182-183). I suggest that those who

¹ For instance, see Alva Noë, *Out of Our Heads: Why You Are Not Your Brain, and Other Lessons from the Biology of Consciousness* (New York: Hill and Wang 2009).

wish to emulate Newberg's bold empirical stance revisit the early history of psychical research, when a group of exceptionally open-minded scientists (such as William James) embarked on a futile search for empirical evidence of the soul.

Such criticisms aside, Newberg's intellectual courage and passion for interdisciplinary research are extraordinary, and I have no doubt he will continue to make significant contributions to the scientific understanding of religion. He is an ideal scientific collaborator for humanistic scholars of religion, the flaws of his proposed 'ideological foundations' notwithstanding. Which brings me to my last point: does interdisciplinary research really need necessary principles or foundations in order to be successful?